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Examples of four elementary functions in R

► [Loading the compressed file \[folder:"ADT_R"\] of source codes and data](#)

The codes of these four functions and their contexts of use are pedagogical examples of the book *Analyse des Données textuelles* [Analysis of Textual Data] (L. Lebart, B. Pincemin, C. Poudat), Presses de l'Université du Québec [in French], 2019, in press.

The R software is both a language and a library. Like Python, it is a free, and multiplatform software. It is the non-proprietary component of the S programming language (Chambers, 1998), itself derived from work of a team of Bell Laboratories dating back to 1976 (team composed of Rick Becker, John Chambers, Doug Dunn, Paul Tukey, and Graham Wilkinson).

1- Compression and reconstruction of images (with graphical display)

2- Principal components analysis (with graphical display)

3- Correspondence analysis (with graphical display)

4- Computation and Plot (on a principal plane) of the Minimum Spanning Tree



1- Compression and reconstruction of images (with graphical display)

This is simply to penetrate, thanks to R, in the black box of a commonplace functionality available in most statistical software of image analysis.

Evidently, the code could be much more compact, but perhaps less readable.

Commands for the R interface

```
#-----  
source("c:\\ADT_R\\svd_image_E.r") # source code  
path= "c:\\ADT_R\\Cheetah.txt"    # data
```

```

X=read.table(path, header=FALSE, row.names=1, sep = "")
nd = 8      # number of axes for the reconstitution
            (e.g.)
Y = recons(X, nd)
visu_image(Y)
#-----

```

The small data-test file "Cheetah.txt" is provided in the folder ADT_R. It is a simple numerical table (200 x 320) [gray levels for each pixel]. See: Nelson M. (1993). *La compression des données. Text, images, sons*. Dunod, Paris.

Case of a more general image in ".pgm" format (Portable Gray Map)

The "Cheetah.txt" file in the example is a simple matrix which is more rustic than the more common image formats (bmp, jpg, png).

Such images must be converted to pgm format by a free software such as GIMP or IrfanView, and the X matrix of command lines is then obtained by the four command lines below:

```

#-----
# Four command lines to substitute to the line:
# "X = read.table ..." in the case of an image in pgm
format
#-----
install.packages ( "pixmap"); # to use only the first
time ...
library (pixmap); # Loading the "pixmap package"
imag = read.pnm ("c:\\ADT_R\\Cheetah.pgm"); # for a
"pgm" image
X = 255 * imag @ gray; # We can now call the function
recons ()
#-----

```

The image below is the image produced by the function recon():

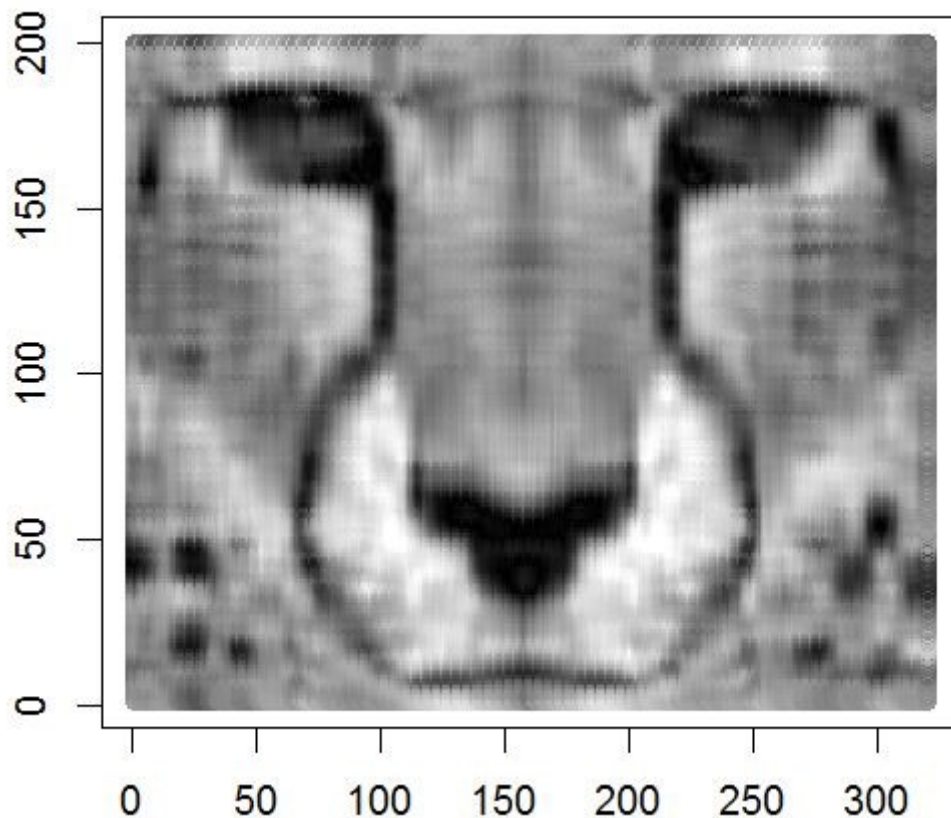


Image obtained after typing the previous R command lines. Compression rate $200/8 = 25$ (8 principal axes).



A2- Principal components analysis (with graphical display)

This is simply to obtain, thanks to R, a functionality available in most statistical software of exploratory multivariate analysis.

Again, the code could be much more compact, but perhaps less readable.

Commands for the R interface

#-----

```

# Commands to be typed in the R interface.
source("c:/ADT_R/ACPS_E.r")           # location of
source code
path= "c:/ADT_R/Mini_Semio.txt"       # location of
data
X=read.table(path, header=TRUE, row.names=1, sep = ",")
# X = data table
Y =ACP(X)                             # main function
plot_ind (Y, X, hor=2, ver=3, font = 1) # graph of
individuals (lines)
                                     # (plane (2,
3) )
plot_var (Y, X, hor=2, ver=3, font = 1) # graph of
variables (columns)
                                     # (plane (2,
3) )
#-----

```

The small data-test file “Mini_semio.txt” is provided in the folder ADT_R. It is a simple numerical table (12 x 7) giving the scores given by 12 respondents to a set of 7 words. The real *semiometric list* contains 210 words, and sample sizes vary from 800 to several thousands. See: *The Semiometric Challenge: Words, Lifestyles and Values* (L. Lebart, J-F. Steiner, M. Piron, J. Wisdom), *L2C, 2014*.  [\(Download the Book - pdf\)](#)  [\(Download the book cover - pdf\)](#) ... and (in French) *La sémiométrie* (L. Lebart, M. Piron, J.-F. Steiner), *Dunod, 2003*.  [\(Download the Book - pdf\)](#)

Small test data table ("Mini-semio.txt")

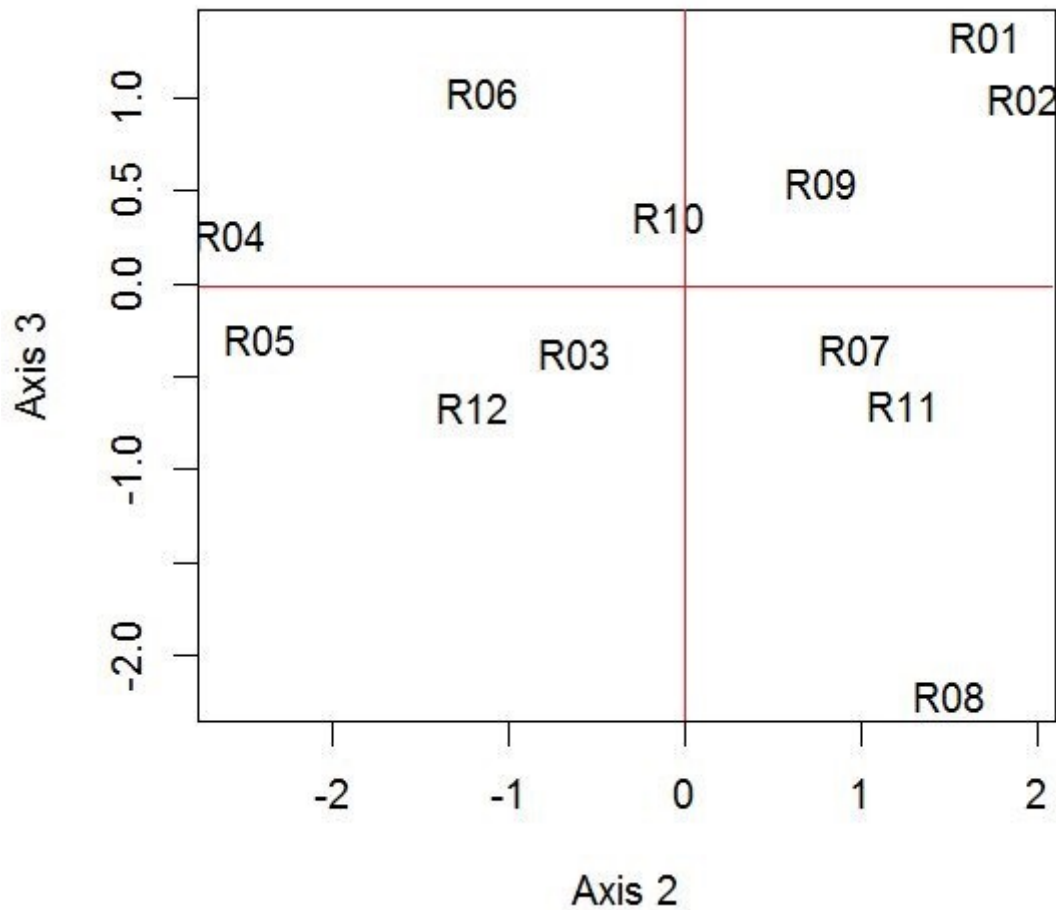
	arbre,	cadeau,	danger,	morale,	orage,	politesse,	sensuel
R01,	7,	4,	2,	2,	3,	1,	6
R02,	6,	3,	1,	2,	4,	1,	7
R03,	4,	5,	3,	4,	3,	4,	3
R04,	5,	5,	1,	7,	2,	7,	1
R05,	4,	5,	2,	7,	1,	6,	2
R06,	5,	7,	1,	5,	2,	6,	5
R07,	4,	2,	1,	3,	5,	3,	6
R08,	4,	1,	5,	4,	5,	4,	7
R09,	6,	6,	2,	4,	7,	5,	5
R10,	6,	6,	3,	5,	3,	6,	6
R11,	7,	7,	6,	7,	7,	6,	7
R12,	2,	2,	1,	2,	1,	3,	2

Meaning of the columns (words of the semiometric list):

arbre = tree;	cadeau = gift;
danger = danger.	morale = moral;
orage = storm;	politesse = politeness;

```
sensuel = sensual.
```

The image below is the image produced by the functions **ACP()** and **plot_ind()**:



**Image obtained after typing the first five R command lines.
Display of the 12 respondents in the plane (2, 3)**



3- Correspondence analysis (with graphical display)

This is a simple and autonomous code for Correspondence Analysis, including a simultaneous representation of rows and columns.

Again, the code could be much more compact, but perhaps less readable.

Commands for the R interface

```
#-----  
# Commands to be typed in the R interface.  
  
#-----  
-----  
# function Y = afcor(X) : Correspondence Analysis of  
matrix X  
#-----  
-----  
# Example for a contingency table "tab.csv" in the  
folder ADT_R  
# (the source code: - afcor_E.r - being in the same  
folder)  
  source("c:/ADT_R/afcor_E.r")           # location of  
source code  
  path = "c:/ADT_R/tab.csv"             # location of data  
# path leads to an Excel(c) file [format csv]  
  X = read.csv(path, row.names = 1, sep = ";") # X =  
data table  
  Y = afcor(X)  
  plot_simult(Y) # simultaneous plot of rows and columns  
of the data table  
#-----  
-----
```

The small data-test file “tab.csv” is provided in the folder ADT_R. It is a simple lexical table (12 x 6) giving the frequencies of 12 words (rows) in the responses of six categories of respondents. to the following question in a sample survey :

“No one can explain why, perhaps because of what they evoke, we find some words pleasant, others unpleasant.

- 1. As far as you are concerned personally, what are the words that you find most pleasant? (Cite as many words as possible).*
- 2. What are the words that you find most unpleasant? (Cite as many words as possible). ”*

This is a case of "Open semiometry" (without closed list of words), cf. Chapter 4 of the already quoted book: *The Semiometric Challenge: Words, Lifestyles and Values* (L. Lebart, J-F. Steiner, M. Piron, J. Wisdom), L2C, 2014.  [\(Download the Book - pdf\)](#)  [\(Download the book cover - pdf\)](#) ... and (in French) *La sémiométrie* (L. Lebart, M. Piron, J.-F. Steiner), Dunod, 2003.  [\(Download the Book - pdf\)](#)

Small test data table ("tab.csv")

Ident	;	Hm30	Hm55	Hp55	Fm30	Fm55	Fp55	<u>Translations</u>
argent	;	11	15	6	14	9	3	money
courage	;	2	3	11	1	3	4	brave
dormir	;	8	3	1	6	2	0	to sleep
livre	;	2	7	1	9	11	11	a book
maman	;	0	3	0	12	11	3	mother
manger	;	9	5	1	10	3	1	to eat
pardon	;	1	2	6	0	5	5	sorry
peinture	;	0	0	1	2	7	3	painting
plaisir	;	10	6	2	11	2	1	pleasure
politesse	;	0	1	5	0	4	8	politeness
repas	;	3	5	4	0	3	1	meal
soleil	;	26	32	23	54	56	42	sun

Meaning of the columns (categories of respondents):

Hm30 = Male, less than 30;

Hm55 = Male, between 30 and 55;

Hp55 = Male, over 55.

Fm30 = Female, less than 30;

Fm55 = Female, between 30 and 55;

Fp55 = Female, over 55.

The image below is the image produced by the functions **afcor()** and **plot_simult()**:

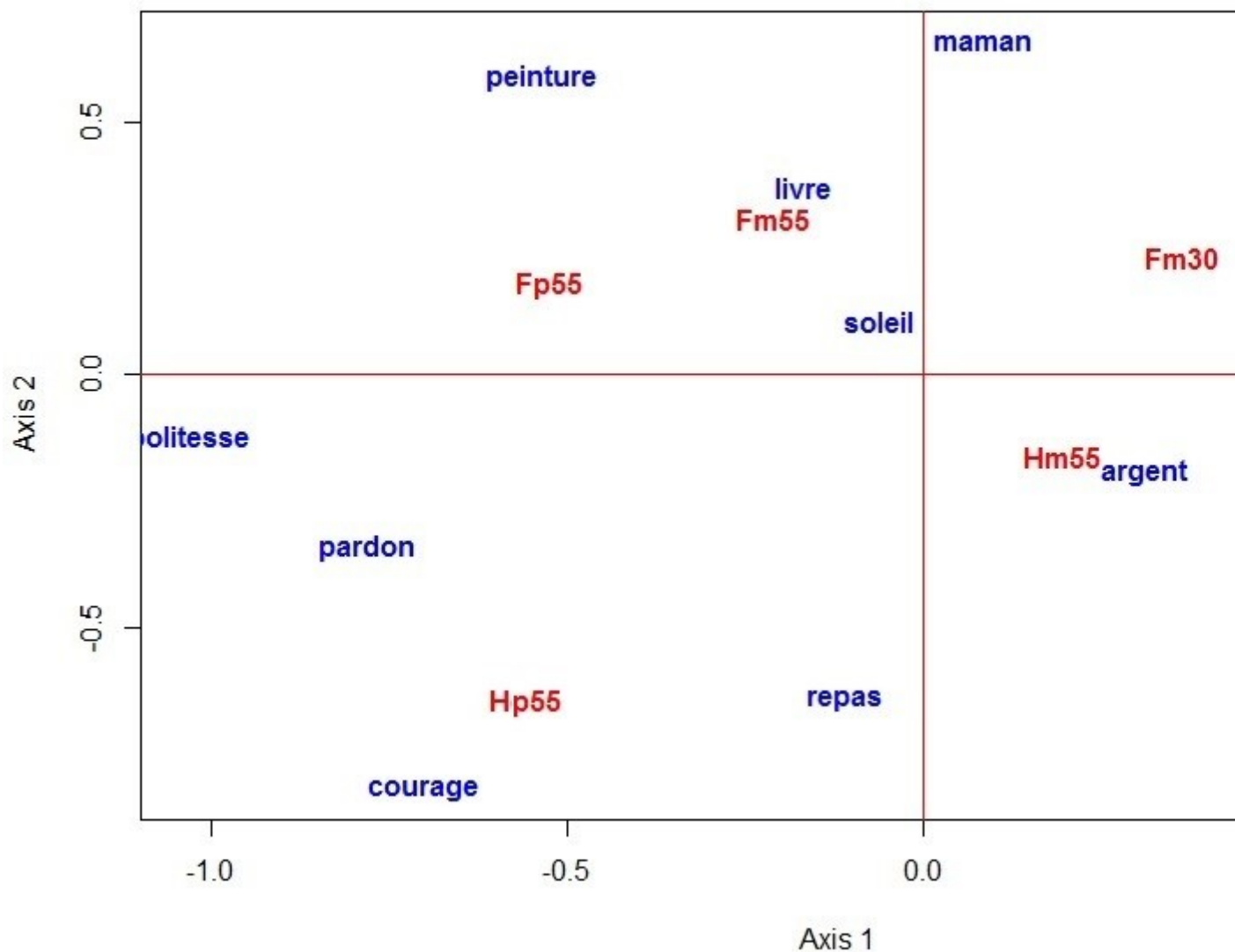


Image obtained after typing the above five R command lines.



A4- Computation and Plot(on a principal plane) of the Minimum Spanning Tree

The basic R language is used again voluntarily without any specialized library. A choice had to be made between the many clustering methods. We chose to implement here the minimum spanning tree (armin function) using the Prim algorithm. The functions print_armin and plot_armin respectively give the list of

the edges of the tree, and a representation of it in a principal plane. The tree is calculated here from the main coordinates resulting from a correspondence analysis (afcor function in section A3 above). With these coordinates, the usual Euclidean distance corresponds to the Chi-2 distance on the initial data.

Finally, a small function afcor_armin_plot (see the end of the listings) provides the desired sequence. It is included in the file armin.r, itself included in the folder titled ADT_R. To obtain the desired results, simply write the following lines:

Commands for the R interface

```
#-----
# Commands to be typed in the R interface.

#-----
# composite function Y = afcor_armin_plot(X) [in the
downloaded folder ADT_R]
#-----
# Example for a contingency table "tab.csv" in the
folder ADT_R
# (the source codes: afcor_E.r and armin_R.r being in
the same folder ADT_R)
source("c:/ADT_R/afcor_E.r")      # location of source
code 1
source("c:/ADT_R/armin_E.r")      # location of source
code 2
path = "c:/ADT_R/tab.csv"         # location of data
# same data as in section A3
afcor_armin_plot(path)
#-----
```

The table below is produced by the function **afcor_armin_plot()**:

Table of vertices obtained after typing the above four R command lines.

NumDeb	edge	NumFin	NomDeb	edge	NomFin
[1,] 1	-----	9	argent	-----	plaisir
[2,] 9	-----	6	plaisir	-----	manger
[3,] 6	-----	3	manger	-----	dormir
[4,] 1	-----	12	argent	-----	soleil
[5,] 12	-----	4	soleil	-----	livre
[6,] 4	-----	5	livre	-----	maman
[7,] 1	-----	11	argent	-----	repas

[8,]	4	-----	8	livre	-----	peinture
[9,]	12	-----	7	soleil	-----	pardon
[10,]	7	-----	10	pardon	-----	politesse

```
NumDeb = Origin Number; NumFin = End Number
NomDeb = Origin Identifier; NomFin = End Identifier
```

The image below is the image produced by the function **afcor_armin_plot()**:

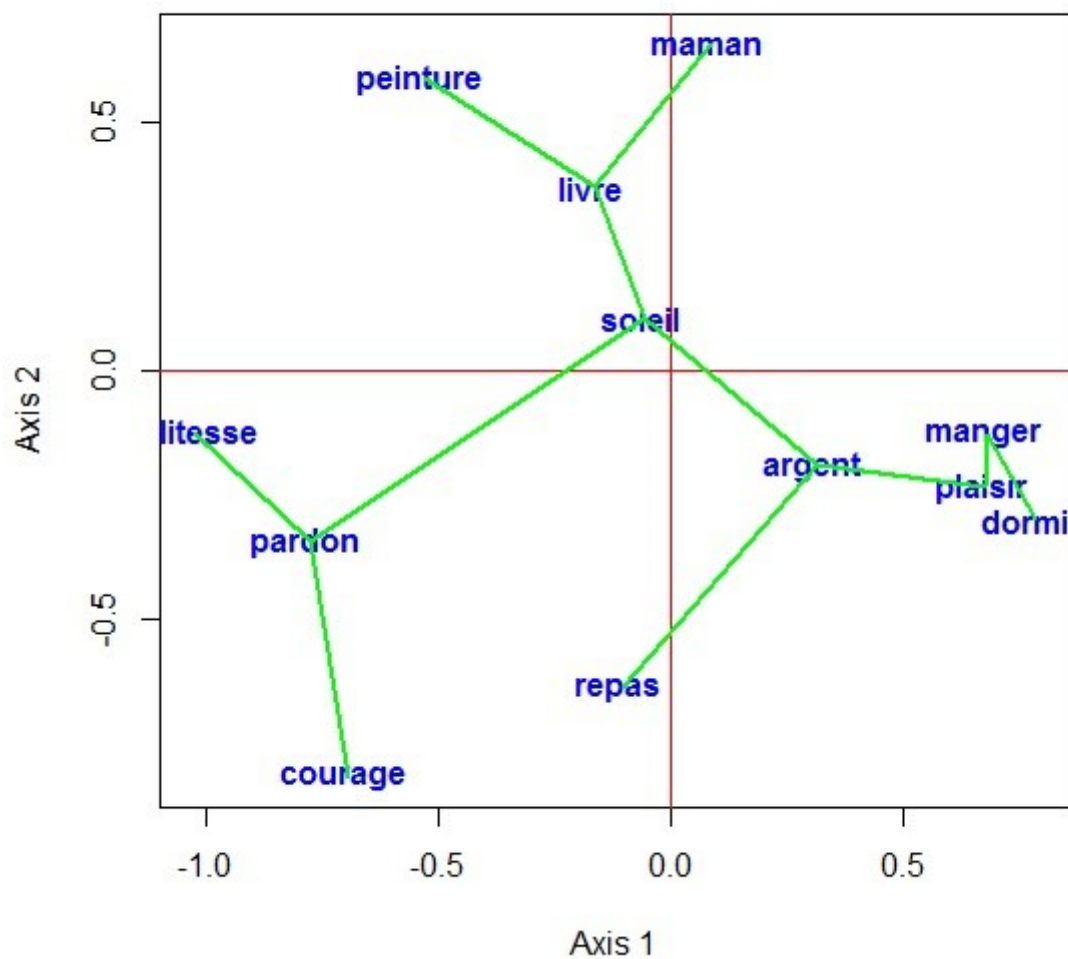


Image obtained after typing the above four R command lines.

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